

1 Week Transformations in Power Query

Work with weeks easily using Power Query!



? What are the options?

There are different options available for **Week**. Those are:

- ✓ Week of Year
- ✓ Week of Month
- ✓ Start of Week
- ✓ End of Week



These options help you extract useful week-related information from a date column.

1.1 Week of Year

Returns the week number within the year.

Date	Week of Year
1/5/2024	1
2/14/2024	7
3/20/2024	12
4/18/2024	16
5/10/2024	19

1.2 Week of Month

Returns the week number within the month.

Date	Week of Month
1/5/2024	1
2/14/2024	3
3/20/2024	3
4/18/2024	3
5/10/2024	2

1.3 Start of Week

Returns the first date of that week.

Date	Start of Week
1/5/2024	1/1/2024
2/14/2024	2/12/2024
3/20/2024	3/18/2024
4/18/2024	4/15/2024
5/10/2024	5/6/2024

1.4 End of Week

Returns the last date of that week.

Date	End of Week
1/5/2024	1/7/2024
2/14/2024	2/18/2024
3/20/2024	3/24/2024
4/18/2024	4/21/2024
5/10/2024	5/12/2024

2 Day Transformations in Power Query

Get detailed day-level insights from your dates!



? What are the options?

There are different options available for **Day**. Those are:

- ✓ Day
- ✓ Day of Week
- ✓ Day of Year
- ✓ Start of Day
- ✓ End of Day
- ✓ Name of Day



These options help you extract useful day-related information from a date column.

2.1 Day

Extracts the day number from the date (e.g., 25).

Date	Day
1/5/2024	5
2/14/2024	14
3/20/2024	20
4/18/2024	18
5/10/2024	10

2.2 Day of Week

Returns which day of the week it is (e.g., Monday = 1).

Date	Day of Week
1/5/2024	6
2/14/2024	3
3/20/2024	4
4/18/2024	4
5/10/2024	6

2.3 Day of Year

Returns the day number within the year (1-365/366).

Date	Day of Year
1/5/2024	5
2/14/2024	45
3/20/2024	80
4/18/2024	109
5/10/2024	131

2.4 Start of Day

Returns the beginning (12:00 AM) of that date.

Date	Start of Day
1/5/2024	1/5/2024 12:00 AM
2/14/2024	2/14/2024 12:00 AM
3/20/2024	3/20/2024 12:00 AM
4/18/2024	4/18/2024 12:00 AM
5/10/2024	5/10/2024 12:00 AM

2.5 End of Day

Returns the end (11:59:59 PM) of that date.

Date	End of Day
1/5/2024	1/5/2024 11:59:59 PM
2/14/2024	2/14/2024 11:59:59 PM
3/20/2024	3/20/2024 11:59:59 PM
4/18/2024	4/18/2024 11:59:59 PM
5/10/2024	5/10/2024 11:59:59 PM

2.6 Name of Day

Returns the name of the weekday (e.g., "Monday").

Date	Name of Day
1/5/2024	Friday
2/14/2024	Wednesday
3/20/2024	Wednesday
4/18/2024	Thursday
5/10/2024	Friday

! Important Note

- ✓ These date functions work directly on a Date column. Make sure your column is in Date data type for accurate results.

✓ Key Takeaways

- ✓ Use Week and Day options to quickly extract date details.
- ✓ Helpful for time-based analysis and reporting.
- ✓ Saves time and reduces manual calculations.

⚙️ Pro Tip

- ✓ Combine these date functions with custom columns to create powerful date dimensions for your reports.



Power BI Desktop

Time Transformations in Power Query

Extract Time Details | Transform with Ease | Better Insights



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Turn Data
Into Better
Decisions

1 What is Time in Power Query?

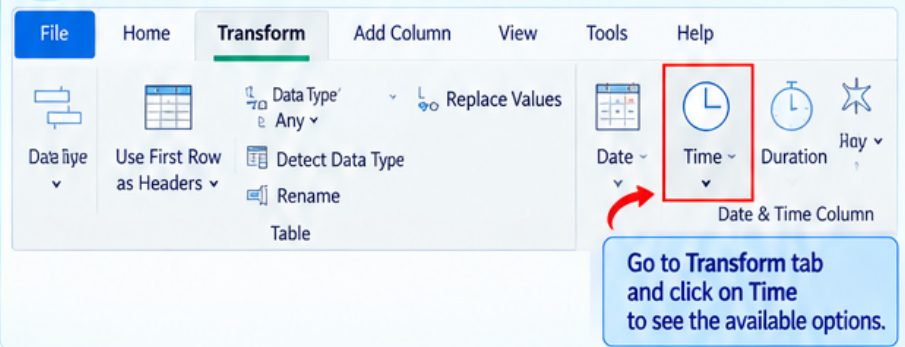
Time is an option in Power Query (under the Transform tab, for time/date-time columns) that lets you extract specific time-related details from a time column, without writing any formula.

There are different options available in the Time menu.

- ✓ Time only
- ✓ Parse
- ✓ Local Time
- ✓ Hour
- ✓ Minute
- ✓ Second
- ✓ Start of Hour / End of Hour



2 Where to Find the Time Option?



Time functions help you easily extract useful time-related information from your data!

*"Small Time Details
Create Big Insights!"*

3 Time Options with Examples

3.1 Time only

Removes the date part and keeps only the time.

Date & Time	Time only
1/5/2024 08:30:15	08:30:15
1/5/2024 14:45:00	14:45:00
1/5/2024 23:10:05	23:10:05

3.2 Parse

Converts text into a proper time format.

Text Value	Parsed Time
08:30:15	08:30:15
14:45:00	14:45:00
23:10:05	23:10:05

3.3 Local Time

Converts the time to local time zone.

UTC Time	Local Time (Example: +5:30)
1/5/2024 08:30:00	1/5/2024 14:00:00
1/5/2024 14:45:00	1/5/2024 20:15:00
1/5/2024 23:10:00	1/6/2024 04:40:00



3.4 Hour

Extracts the hour part from the time (0–23).

Date & Time	Hour
1/5/2024 08:30:15	8
1/5/2024 14:45:00	14
1/5/2024 23:10:05	23

3.5 Minute

Extracts the minute part from the time (0–59).

Date & Time	Minute
1/5/2024 08:30:15	30
1/5/2024 14:45:00	45
1/5/2024 23:10:05	10

3.6 Second

Extracts the second part from the time (0–59).

Date & Time	Second
1/5/2024 08:30:15	15
1/5/2024 14:45:00	0
1/5/2024 23:10:05	5

3.7 Start of Hour / End of Hour

Returns the beginning or end of that hour.

Date & Time	Start of Hour	End of Hour
1/5/2024 08:30:15	08:00:00	08:59:59
1/5/2024 14:45:00	14:00:00	14:59:59
1/5/2024 23:10:05	23:00:00	23:59:59

Key Takeaways

- ✓ Use Time options to quickly extract time-related details.
- ✓ No need to write complex formulas.
- ✓ Helps in data transformation and analysis.
- ✓ Useful for reports, dashboards, and time-based calculations.



Real-World Use Cases

- 📞 Analyze call center data (call time, duration).
- 📅 Work with transaction logs.
- 🕒 Convert UTC time to local time.
- 🔍 Extract hour/minute/second for trend analysis.
- 🏭 Useful in manufacturing, IT logs, and business operations.



Pro Tip

- ✓ Ensure your column is in Time or Date/Time data type.
- ✓ Use Parse when your time is in text format.
- ✓ Use Local Time to adjust time zones.
- ✓ Combine time functions with date functions for deeper analysis.
- ✓ Preview results in Power Query before applying changes.



Power BI

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Microsoft Power BI Desktop

Add Column Tab in Power Query

Create | Transform | Enrich | Get More Insights



AVD Group

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1 Introduction

The **Add Column** tab in Power Query lets you create new columns in your table based on existing data — either using built-in options (like index, custom formulas, or conditional logic) or by extracting/combining data from other columns. It helps you add calculated or derived data without changing the original source.



"Add new insights to your data with just a few clicks!"

2 Options under Add Column Tab



Column From Example
(Create column using examples)



Custom Column
(Write your own formula)



Conditional Column
(Add logic-based column)

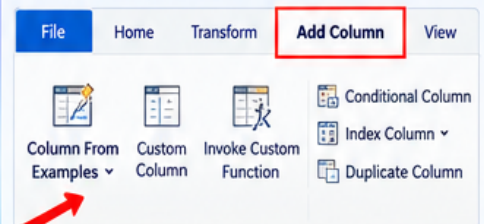


Index Column
(Add index numbers)



Duplicate Column
(Duplicate an existing column)

3 Where to Find Add Column Tab?



Go to **Add Column** tab in Power Query to see all available options.

4 Column From Example

Column From Examples lets you create a new column by simply typing a few example values of what you want the output to look like. Power Query automatically detects the pattern and generates the required formula/logic — so you don't need to write any code yourself.

Two Options are Available:



From all Columns
(Use any/all columns)



From Selection
(Use selected column)

5 From all Columns

Lets you create a new column by giving examples based on data from any/all columns in the table. Power Query looks across all columns to detect the pattern.

This option is used in 2 ways:

- 1 Combine multiple columns into single column.
- 2 Using existing columns in the table we can generate default options.

6 Combine multiple columns into single column

If you want to combine multiple columns into a single column, you can use the 'From All Columns' option.

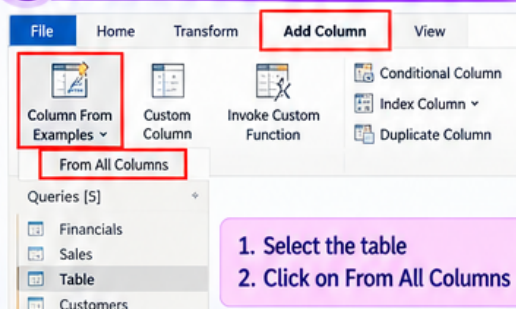
Steps:

- 1 Select the table.
- 2 Go to **Add Column** → click on **From All Columns**.
- 3 Fill the example values.
- 4 Click **OK**.

Example Output

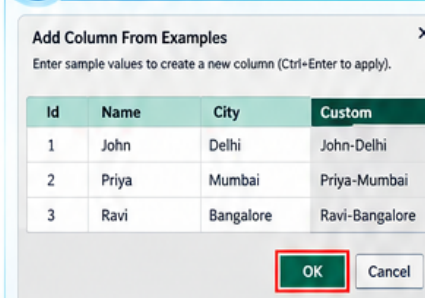
Combine multiple columns like Name + City into a single column easily!

7 Step 1: Select Table and Click on From All Columns



1. Select the table
2. Click on From All Columns

8 Step 2: Fill Example Values



9 Result: New Column Generated

Id	Name	City	Merged
1	John	Delhi	John-Delhi
2	Priya	Mumbai	Priya-Mumbai
3	Ravi	Bangalore	Ravi-Bangalore



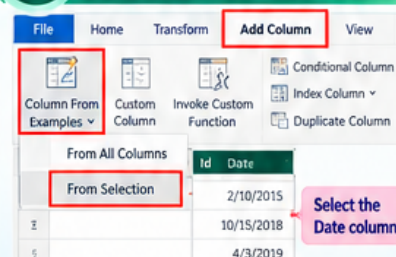
A new column will be generated with the combined values from all the selected columns.

10 From Selection (Using Existing Column)

You can select any column and click on 'From Selection'; the default options will appear, and you can choose whichever option you require.

For example, select the **Date** column, click on 'From Selection', and follow the steps.

11 Example: Select Date Column



12 Result: Choose Option and Create Column

After selecting 'From Selection', fill the example values (or choose from the suggested options). A new column will be automatically created based on the selected option.



Quickly generate date-based columns like Year, Month, Day, etc.



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Custom Column & Conditional Column in Power Query

Create new columns | Apply logic | Transform data with ease!

"Small Steps
Big Insights!"

1

Custom Column

Create new columns using your own M language formula



What is Custom Column?

Custom Column is an option in Power Query that lets you create a new column by writing your own simple formula, using the values from other columns in your table. For example, you can create a "Profit" column by writing a formula like Sales minus Cost, and Power Query will calculate it for every row automatically.

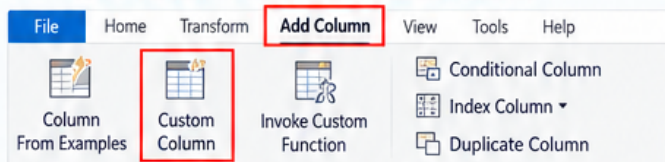
Example: Generate Month Name from Date Column

Let's create a Month Name column from an existing Date column using an M Language formula.

1

Click on Custom Column

Go to **Add Column** tab and click on **Custom Column**.



2

Write the M Language Formula

Provide a new column name and write the formula.

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ①

Available columns

- Id
- Address
- Date
- Merged
- Month Name

<< Insert

✓ No syntax errors have been detected.

OK Cancel

3

Result: Month Name Column is Generated

A new column will be added to the table with the Month Name.

Id	Address	Date	Month Name
1	3-22,Kantam	2/10/2015	February
2	4-21,Poduru	10/15/2008	October
3	5-23,Psam	4/3/2019	April
4	6-21,Onavo	7/21/2012	July
5	2-12,Karu	12/25/2022	December
6	1-21,Ovalu	8/17/2017	August



Key Takeaways

- ✓ Custom Column uses M language formulas.
- ✓ Conditional Column uses simple UI (no coding).
- ✓ Helps create calculated and derived columns.
- ✓ Useful for data transformation and analysis.



Useful Link

Explore all M functions here:



<https://learn.microsoft.com/en-us/powerquery-m/power-query-m-function-reference>



Pro Tips

- ✓ Use meaningful column names.
- ✓ Verify results before applying changes.
- ✓ Use Custom Column for complex logic.
- ✓ Use Conditional Column for simple conditions.
- ✓ These options save time and make your queries powerful!

2

Conditional Column

Create new columns using IF-THEN-ELSE logic (no coding required)



What is Conditional Column?

Conditional Column is an option in Power Query (under the Add Column tab) that lets you create a new column based on **IF-THEN-ELSE** logic, without writing any M formula manually. You set up conditions (like "if Sales > 1000, then 'High', else 'Low'"), and Power Query automatically generates the required column using a simple interface.

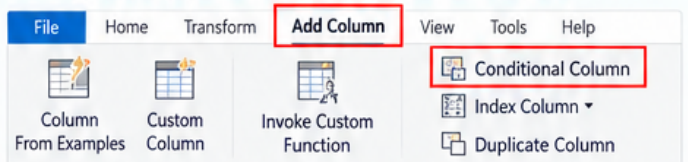
Example: Generate Month Number from Month Name

Suppose you have a Month Name column, and using that, you want to generate a Month Number column with Conditional Column.

1

Click on Conditional Column

Go to **Add Column** tab and click on **Conditional Column**.



2

Define If-Else If-Else Conditions

Provide a new column name and set the conditions accordingly.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

If

Month Name equals January Then 01

Else If

Month Name equals February Then 02

Else If

Month Name equals March Then 03

Else If

Month Name equals April Then 04

Else If

Month Name equals May Then 05

Else If

Month Name equals June Then 06

Add Clause

Else ①

if 12

OK Cancel

3

Result: Month Number Column is Generated

A new column will be added to the table with the Month Number.

Id	Date	Month Name	Month No
1	2/10/2015	February	2
2	10/15/2008	October	10
3	4/3/2019	April	4
4	7/21/2012	July	7
5	12/25/2022	December	12
6	8/17/2017	August	8



Index Column & Duplicate Column in Power Query

Simple Options. Powerful Results.

Small Steps
Big Insights!

1 Index Column



Add sequential numbers to your table easily!



What is Index Column?

Index Column is an option in Power Query (under the Add Column tab) that lets you add a new column with sequential numbers (like a row ID) to your table — for example, 1, 2, 3, 4... or starting from 0. It's useful for uniquely identifying each row or tracking the original row order.

There are 3 options available:



From 0



From 1



Custom Column

1.1 From 0

Adds a new column starting the numbering from 0 (0, 1, 2, 3...).

Data	Index
A	0
B	1
C	2
D	3

1.2 From 1

Adds a new column starting the numbering from 1 (1, 2, 3, 4...).

Data	Index
A	1
B	2
C	3
D	4

1.3 Custom Column

Lets you set your own starting number and increment value (e.g., start from 100, increase by 5 → 100, 105, 110...).

Data	Index
A	100
B	105
C	110
D	115

Key Takeaways – Index Column

- ✓ Use to add row numbers.
- ✓ Choose From 0, From 1, or Custom.
- ✓ Helpful for unique identification and tracking row order.
- ✓ Very useful in data transformation and analysis.



2 Duplicate Column



Create an exact copy of an existing column!



What is Duplicate Column?

Duplicate Column is an option in Power Query (under the Add Column tab) that lets you create an exact copy of an existing column, with the same data, placed right next to the original column. This is useful when you want to keep the original column unchanged while applying different transformations on the copy.

There are 2 ways to create Duplicate Column:

- 1 Select the existing column and click on **Duplicate**.
- 2 Right click on any existing column and click on **Duplicate Column**.

2.1 Method 1: Using Add Column Tab

Select the existing column, then go to **Add Column** tab and click on **Duplicate Column**.

Id	Name	Month Name	Month Name - Copy
1	John	January	January
2	Priya	February	February
3	Ravi	March	March
4	Sita	April	April

2.2 Method 2: Using Right Click

Right click on any existing column and click on **Duplicate Column**.

Key Takeaways – Duplicate Column

- ✓ Creates an exact copy of an existing column.
- ✓ Helps keep the original data unchanged.
- ✓ Can be created using the Add Column tab or Right Click.
- ✓ Useful for applying different transformations on the copy.



Clean Data
Shape Data
Create Better Insights!

Power Query (Transform Data)

Home | Transform | Add Column

Small Steps
Big Impact!



Home Tab



Close & Apply

- Close & Apply
- Apply
- Close



New Source



Recent Sources



Enter Data



Data Source Settings

- Change Source
- Edit Permissions
- Clear Permissions
- Clear all Permissions



Manage Parameters



Refresh Preview

- Refresh Preview
- Refresh All



Properties



Applied Steps



Advanced Editor



Manage

- Delete
- Duplicate
- Reference



Choose Columns

- Choose Columns
- Goto Columns



Remove Columns

- Remove Columns
- Remove Other Columns



Remove Rows

- Remove Duplicates
- Remove Errors
- Remove Blank Rows



Sort



Split Column

- By Delimiter



Group By



Data Type

- Decimal Number
- Whole Number
- Percentage
- Date/Time
- Date
- Time
- Text
- True/False



Use First Row as Headers

- Use First Row as Headers
- Use Headers as First Row



Replace



Merge Queries



Append Queries

- Two Tables
- Three or More Tables



Transform Tab



Group By



Use First Row as Headers



Reverse Rows



Count Rows



Data Type



Detect Data Type



Rename



Replace Values

- Replace Values
- Replace Errors



Pivot Column



Un Pivot Columns

- Unpivot Columns
- Unpivot Other Columns
- Unpivot Only Selected Columns



Move

- Left
- Right
- To Beginning
- To End



Format

- Lowercase
- UPPERCASE
- Capitalize Each Word
- Trim
- Clean
- Add Prefix
- Add Suffix



Merge Columns



Extract

- Length
- First Characters
- Last Characters
- Range
- Text Before Delimiter
- Text After Delimiter
- Text Between Delimiters



Standard

- Add
- Multiply
- Subtract
- Divide
- Modulo
- Percentage
- Percentage of



Rounding

- Round
- Round Up
- Round Down



Information

- Is Even
- Is Odd
- Sign



Date

- Age
- Date Only
- Parse
- Earliest
- Latest



Year

- Year
- Start of Year
- End of Year



Month

- Month
- Start of Month
- End of Month
- Days in Month
- Name of Month



Quarter

- Quarter of Year
- Start of Quarter
- End of Quarter



Week

- Week of Year
- Week of Month
- Start of Week
- End of Week



Day

- Day
- Day of Week
- Day of Year
- Start of Day
- End of Day
- Name of Day



Combine Date and Time



Time

- Time only
- Parse
- Local Time
- Hour
- Minute
- Second
- Start of Hour
- End of Hour



Add Column Tab



Column From Example

- From all Columns
- From Selection



Custom Column



[Click here for
M-Language Functions
Reference](#)



Conditional Column



Index Column

- From 0
- From 1
- Custom Column



Duplicate Column

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Transform Data Today for a Smarter Tomorrow!